Nathaniel Mann

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Third-year Electrical Engineering student and Queen's varsity athlete with hands-on experience in embedded systems, Python automation, data visualization, CAD and electronics prototyping. Experience developing integrated hardware/software systems and leading multidisciplinary technical projects.

EDUCATION

Bachelor of Applied Science, Queen's University, Kingston, ON | Expected 2028

Electrical and Computer Engineering – Innovation (ECE-I) specialty

- Coursework: Computer Networks, Computer Programming 2, ECE Design & Practice, Data Science

WORK EXPERIENCE

Business Analyst - Regional Animal Protection Society (RAPS) Jun-Sept 2026

- Automated an export of 1,700+ client records using Python and Selenium WebDriver, to complete an estimated manual-entry 2-day job in approximately 40 minutes.
- Developed an interactive Power BI dashboard (Python, DAX) with geospatial client heatmaps and custom Matplotlib gauge charts, presented monthly to the CEO and 11-member board.
- Led a 6-person team in the design and development of an online auction platform for Mama Karen's Korner, from initial proposal through Shopify implementation and project handoff.
- Analyzed drug costs by dog and cat size and prior-year usage across 12 monthly specials in Python and Excel to identify the cost-effective options for the RAPS Animal Hospital.
- 3D-Printed 3 wheelchairs for kittens with cerebellar hypoplasia (wobbly kittens).

Labourer - Coastline Projects Ltd. May-Sept 2025

- Supported residential construction projects in a hands-on team environment.

Project Support - ECOmetrix Environmental Consulting Inc. 2024

- Supported the development of a literature-based assessment of Marine Protected Areas in Canada.

PROJECTS

Lettuce Investment Tool 2026

- Built and deployed Lettuce (app.nathanielmann.ca), a stock research app for any SEC-filing U.S. company, using Node.js and vanilla JS with zero npm dependencies (server, SVG charts, tests).
- Calculates fair value, P/E, ROIC, CAPM/WACC (60-month beta) and a two-stage, 10-year DCF from raw SEC EDGAR XBRL filings, 20+ years of Cboe price history and FRED Treasury rates, with no API keys.
- Wrote an XBRL pipeline that auto-detects stock splits and derives Q4 and TTM EPS, with a 24-hour server cache, a 60-load/hour rate limit and 30 automated tests gating every deploy.

1995 Toyota 4Runner 2026

- Diagnosed power window failures using a battery and multimeter to isolate a wiring fault and tracing another failure to a rusted arm, which was replaced per manufacturer service specifications
- Diagnosed and replaced the rear brake lines, restoring full braking performance.

Battery Management System for QRET Avionics 2025

- Assembled and voltage-tested battery pack layouts for QRET avionics and researched BMS components using circuit board datasheets.

TECHNICAL SKILLS

- **Data Analysis:** NumPy, pandas, matplotlib, scikit-learn, Excel, Power BI, Power Query
- **Engineering & Design**: SolidWorks, Quartus 2, KiCad 8 & 9, Arduino, ESP32, soldering, 3D-Printing, Codex, Claude Code, git/GitHub, APIs
- **Languages**: Proficient: C, Assembly, Python | Familiar: C#, C++, Java, JavaScript, VHDL, DAX, HTML, SQL

CAMPUS ACTIVITIES

Queen's Rocket Engineering Team – Avionics Sub-team general member 2025-Present
Queen's Varsity Men's Ultimate Team 2024-Present
Queen's University Motionball Athlete Ambassador 2025-Present